

Feasibility Study

PhishShield

A Machine Learning-Based Web
Application for Phishing URL Detection

GROUP MEMBERS:

- ABHIJITH BIJU
- AMEENA MARYAM
- RIFANA RASHEED
- VISWAS VIVEK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COLLEGE OF ENGINEERING POONJAR

Abstract

Phishing attacks are one of the most common cybersecurity threats, where attackers create fraudulent websites to steal sensitive information such as usernames, passwords, banking credentials, and personal data. Traditional blacklist-based detection methods are often unable to identify newly created phishing websites. To address this issue, the proposed project, PhishShield, aims to develop a machine learning-based web application that analyzes URL features and predicts whether a website is legitimate or phishing in real time. This feasibility study evaluates the practicality of the project from technical, economic, operational, and schedule perspectives and confirms that the project can be successfully implemented within the available resources and academic timeline.

Keywords: *phishing detection, machine learning, feasibility study, web application, cybersecurity, URL classification*

1. Executive Summary

This feasibility study evaluates the practicality of developing PhishShield: A Machine Learning-Based Web Application for Phishing URL Detection. The study examines the project's technical, economic, operational, and schedule feasibility.

The proposed system uses machine learning techniques to analyze URL-based features and predict whether a website is legitimate or phishing. The analysis shows that the project can be developed using open-source technologies, publicly available datasets, and standard development tools with minimal cost. It is user-friendly, practical to implement, and can be completed within the academic project timeline.

Based on the feasibility analysis, PhishShield is considered a viable and effective solution for real-time phishing URL detection and improving cybersecurity awareness.

2. Project Description

PhishShield is a web-based application designed to detect phishing URLs using machine learning techniques. The system allows users to enter a website URL, extracts various URL-based features, and uses a trained machine learning model to classify the URL as either Legitimate or Phishing. The application provides users with immediate prediction results, helping them avoid fraudulent websites and improving cybersecurity awareness.

The project combines web technologies with machine learning to deliver a lightweight, accessible, and intelligent phishing detection solution.

3. Technical Feasibility

The proposed system is technically feasible because it is developed using widely accepted and open-source technologies. The backend will be implemented using Python and the Flask framework, while the frontend will use HTML, CSS, and JavaScript to provide a simple and interactive user interface. The phishing detection model will be developed using Scikit-learn and trained on publicly available phishing URL datasets.

The required software tools, including Visual Studio Code, GitHub, Python, and machine learning libraries such as Pandas, NumPy, and Scikit-learn, are freely available. The project can be developed and tested on a standard laptop without requiring specialized hardware. Therefore, the proposed system is technically practical and can be successfully implemented using the available resources.

4. Economic Feasibility

The proposed project is economically feasible because it requires very little financial investment. The programming languages, development frameworks, machine learning libraries, and datasets are open source and freely available. Existing personal computers can be used for development and testing, eliminating the need for additional hardware expenses.

If the application is deployed online, free cloud hosting services can be used during development and demonstration. Since no commercial software licenses or expensive infrastructure are required, the total development cost remains low, making the project highly suitable for academic implementation.

5. Operational Feasibility

The proposed web application is simple, user-friendly, and easy to operate. Users only need to enter a website URL into the application, and the system automatically analyzes the URL and predicts whether it is legitimate or phishing. The application displays the prediction

along with a security recommendation, enabling users to make informed decisions before accessing websites.

The system is intended for students, employees, businesses, and general internet users. Since it requires minimal user interaction and no technical expertise, the application is expected to achieve high user acceptance while contributing to increased cybersecurity awareness.

6. Schedule Feasibility

The project is feasible within the allotted academic project schedule. The development process has been divided into manageable phases, including project planning, requirement analysis, literature survey, dataset collection, feature extraction, machine learning model development, web application development, testing, documentation, and final deployment.

Proper work allocation among team members and regular progress monitoring through GitHub will ensure that each milestone is completed on time. The availability of open-source tools and existing datasets further reduces development time, making it practical to complete the project within the semester.

7. Findings and Recommendation

The feasibility analysis indicates that the proposed PhishShield: A Machine Learning-Based Web Application for Phishing URL Detection is technically, economically, operationally, and schedule feasible. The project utilizes readily available technologies, requires minimal development cost, offers a simple and effective user experience, and can be completed within the academic timeline. Therefore, the implementation of the proposed system is recommended as it provides a practical and intelligent solution for real-time phishing URL detection while promoting cybersecurity awareness.